analytically; this is explored in one of the exercises at the end this chapter.

A brief history of the evolution of robot dynamics algorithms, as well as pointers to references on the more general subject of multibody system dynamics (of which robot dynamics can be considered a subfield) can be found in [48].

8.13 Exercises

Exercise 8.1 Derive the formulas given in Figure 8.5 for:
(a) a rectangular parallelpiped;
(b) a circular cylinder;
(c) an ellipsoid.

Exercise 8.2 Consider a cast iron dumbbell consisting of a cylinder connecting two solid spheres at either end of the cylinder. The density of the dumbbell is 7500 kg/m^3. The cylinder has a diameter of 4 cm and a length of 20 cm. Each sphere has a diameter of 20 cm.
(a) Find the approximate rotational inertia matrix $\mathcal{I}_b$ in a frame {b} at the center of mass with axes aligned with the principal axes of inertia of the dumbbell.
(b) Write down the spatial inertia matrix $\mathcal{G}_b$.

Exercise 8.3 Rigid-body dynamics in an arbitrary frame.
(a) Show that Equation (8.42) is a generalization of Steiner's theorem.
(b) Derive Equation (8.43).

Exercise 8.4 The 2R open-chain robot of Figure 8.16, referred to as a rotational inverted pendulum or Furuta pendulum, is shown in its zero position. Assuming that the mass of each link is concentrated at the tip and neglecting its thickness, the robot can be modeled as shown in the right-hand figure. Assume that $\mathfrak{m}_1 = \mathfrak{m}_2 = 2$, $L_1 = L_2 = 1$, $g = 10$, and the link inertias $\mathcal{I}_1$ and $\mathcal{I}_2$ (expressed in their respective link frames $\{b_1\}$ and $\{b_2\}$) are

$$\mathcal{I}_1 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}, \qquad \mathcal{I}_2 = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

(a) Derive the dynamic equations and determine the input torques τ_1 and τ_2 when $\theta_1 = \theta_2 = \pi/4$ and the joint velocities and accelerations are all zero.

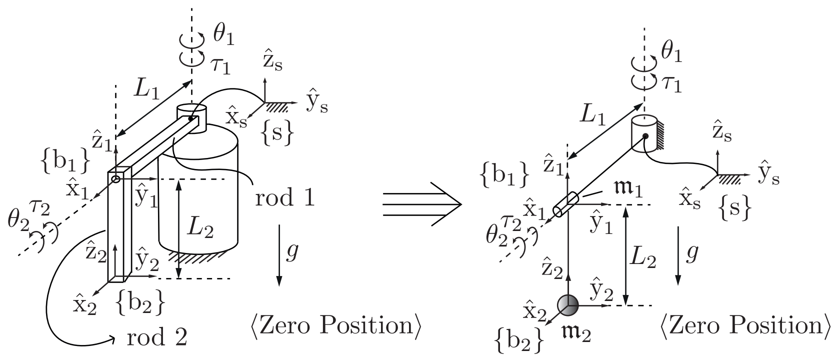

Figure 8.16: 2R rotational inverted pendulum. (Left) Its construction; (right) the model.

(b) Draw the torque ellipsoid for the mass matrix $M(\theta)$ when $\theta_1 = \theta_2 = \pi/4$.

Exercise 8.5 Prove the following Lie bracket identity (called the Jacobi identity) for arbitrary twists $\mathcal{V}_1$, $\mathcal{V}_2$, $\mathcal{V}_3$:

$$\mathrm{ad}_{\mathcal{V}_1}(\mathrm{ad}_{\mathcal{V}_2}(\mathcal{V}_3)) + \mathrm{ad}_{\mathcal{V}_3}(\mathrm{ad}_{\mathcal{V}_1}(\mathcal{V}_2)) + \mathrm{ad}_{\mathcal{V}_2}(\mathrm{ad}_{\mathcal{V}_3}(\mathcal{V}_1)) = 0.$$

Exercise 8.6 The evaluation of $\dot{J}(\theta)$, the time derivative of the forward kinematics Jacobian, is needed in the calculation of the frame accelerations $\dot{\mathcal{V}}_i$ in the Newton–Euler inverse dynamics algorithm and also in the formulation of the task-space dynamics. Letting $J_i(\theta)$ denote the ith column of $J(\theta)$, we have

$$\frac{d}{dt} J_i(\theta) = \sum_{j=1}^{n} \frac{\partial J_i}{\partial \theta_j} \dot{\theta}_j.$$

(a) Suppose that $J(\theta)$ is the space Jacobian. Show that

$$\frac{\partial J_i}{\partial \theta_j} = \begin{cases} \mathrm{ad}_{J_j}(J_i) & \text{for } i > j \\ 0 & \text{for } i \leq j. \end{cases}$$

(b) Now suppose that $J(\theta)$ is the body Jacobian. Show that

$$\frac{\partial J_i}{\partial \theta_j} = \begin{cases} \mathrm{ad}_{J_i}(J_j) & \text{for } i < j \\ 0 & \text{for } i \geq j. \end{cases}$$

Exercise 8.7 Show that the time derivative of the mass matrix $\dot{M}(\theta)$ can be written explicitly as

$$\dot{M} = -\mathcal{A}^{\mathrm{T}}\mathcal{L}^{\mathrm{T}}\mathcal{W}^{\mathrm{T}}[\mathrm{ad}_{\mathcal{A}\dot{\theta}}]^{\mathrm{T}}\mathcal{L}^{\mathrm{T}}\mathcal{G}\mathcal{L}\mathcal{A} - \mathcal{A}^{\mathrm{T}}\mathcal{L}^{\mathrm{T}}\mathcal{G}\mathcal{L}[\mathrm{ad}_{\mathcal{A}\dot{\theta}}]\mathcal{W}\mathcal{L}\mathcal{A},$$

with the matrices as defined in the closed-form dynamics formulation.

Exercise 8.8 Explain intuitively the shapes of the end-effector force ellipsoids in Figure 8.4 on the basis of the point masses and the Jacobians.

Exercise 8.9 Consider a motor with rotor inertia $\mathcal{I}_{\mathrm{rotor}}$ connected through a gearhead of gear ratio G to a load with scalar inertia $\mathcal{I}_{\mathrm{link}}$ about the rotation axis. The load and motor are said to be **inertia matched** if, for any given torque τ_m at the motor, the acceleration of the load is maximized. The acceleration of the load can be written

$$\ddot{\theta} = \frac{G\tau_m}{\mathcal{I}_{\mathrm{link}} + G^2\mathcal{I}_{\mathrm{rotor}}}.$$

Solve for the inertia-matching gear ratio $\sqrt{\mathcal{I}_{\mathrm{link}}/\mathcal{I}_{\mathrm{rotor}}}$ by solving $d\ddot{\theta}/dG = 0$.

Exercise 8.10 Give the steps that rearrange Equation (8.99) to get Equation (8.101). Remember that $P(\theta)$ is not full rank and cannot be inverted.

Exercise 8.11 Program a function to calculate $h(\theta, \dot{\theta}) = c(\theta, \dot{\theta}) + g(\theta)$ efficiently using Newton–Euler inverse dynamics.

Exercise 8.12 Give the equations that would convert the joint and link descriptions in a robot's URDF file to the data `Mlist`, `Glist`, and `Slist`, suitable for using with the Newton–Euler algorithm `InverseDynamicsTrajectory`.

Exercise 8.13 The efficient evaluation of $M(\theta)$.

(a) Develop conceptually a computationally efficient algorithm for determining the mass matrix $M(\theta)$ using Equation (8.57).

(b) Implement this algorithm.

Exercise 8.14 The function `InverseDynamicsTrajectory` requires the user to enter not only a time sequence of joint variables `thetamat` but also a time sequence of joint velocities `dthetamat` and accelerations `ddthetamat`. Instead, the function could use numerical differencing to find approximately the joint velocities and accelerations at each timestep, using only `thetamat`. Write an alternative `InverseDynamicsTrajectory` function that does not require the user to enter `dthetamat` and `ddthetamat`. Verify that it yields similar results.

Exercise 8.15 Dynamics of the UR5 robot.

(a) Write the spatial inertia matrices $\mathcal{G}_i$ of the six links of the UR5, given the center-of-mass frames and mass and inertial properties defined in the URDF in Section 4.2.

(b) Simulate the UR5 falling under gravity with acceleration $g = 9.81$ m/s^2 in the $-\hat{\mathrm{z}}_{\mathrm{s}}$-direction. The robot starts at its zero configuration and zero joint torques are applied. Simulate the motion for three seconds, with at least 100 integration steps per second. (Ignore the effects of friction and the geared rotors.)